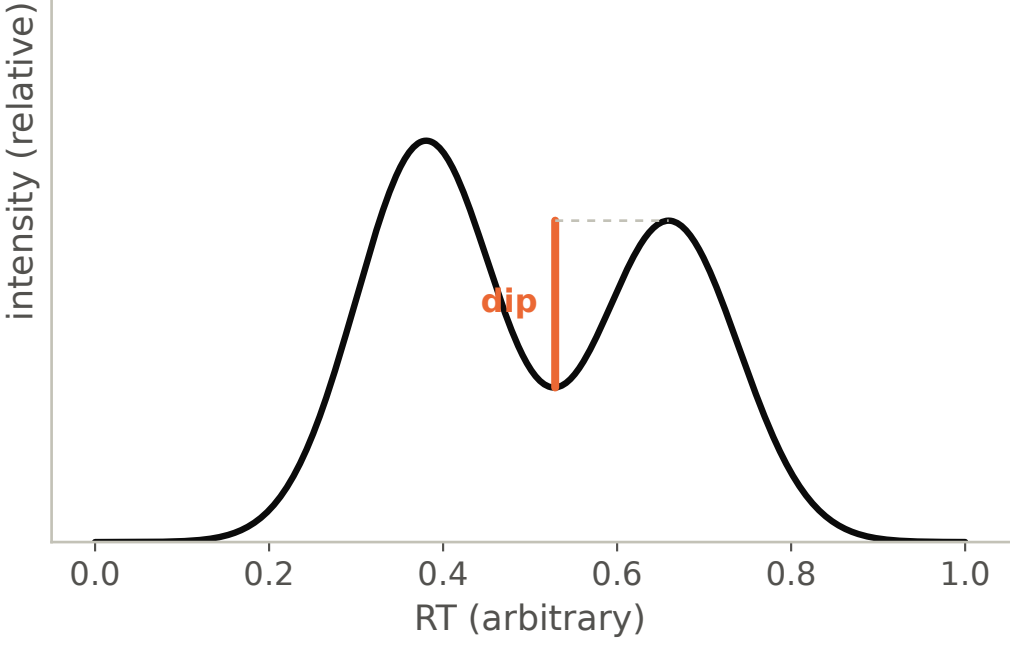
a dip counts as a valley when it exceeds 3 σ of the two levels, σ from the law:

weak ion, apex 40k: dip 16.7k vs 3 σ = 36.8k → ripple, no valley
strong ion, apex 4M: dip 1.67M vs 3 σ = 457.7k → a real valley